

# TOBY CRUST

SOFTWARE ENGINEER — AI WORKFLOWS • FULL-STACK & INTEGRATIONS • REAL-TIME SYSTEMS

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Software engineer operating across **full-stack web applications** (React, TypeScript), **AI workflow automation** (Python, AI agents, Slack API), and **real-time systems** (Unity, C#). Three years of shipping across SaaS, government, and high-traffic platforms owning products and custom internal tools end-to-end from prototype to production release.

## SELECTED WORK

SLIK • AI & INTERNAL TOOLING

### Slack-to-VM AI Agent & Testing Pipeline

Built an internal AI automation pipeline integrated with Slack to accelerate development and non-dev workflows. Non-technical team members specify requested project changes directly in Slack, which dispatches to an AI agent that provisions a virtual machine, applies code changes, executes test suites, and alerts me for final PR review and merge. Drastically cut routine work and accelerated overall team shipping velocity.

Python AI Agents Slack API VM Automation GitHub Actions Automated Testing CI/CD

SLIK • NORTHERN TERRITORY GOVERNMENT

### Augmented reality heritage app

Shipped a historical WWII AR app for the NT Government on iOS and Android, triggering on-site content via BLE beacons and geolocation. Built with Unity Addressables to reduce app store download size and enable modular, dynamic content updates for future expansion. Further locations and experiences are in production now, due to success of initial app.

Unity C# AR Foundation ARKit / ARCore BLE beacons Addressables Localization Convai

SLIK • FIGHTMND

### Donation platform

Full-stack build of the donation site for a national fundraising campaign — payment flow, supporter data capture and a front end that held up under broadcast-driven traffic spikes.

React TypeScript Stripe Netlify

**\$165,575**

RAISED THROUGH THE PLATFORM

**8.9K**

PEAK CONCURRENT VIEWERS

Dec 2025 – present

**Slik** — Software Engineer, Sydney

- Lead developer on the NT Government AR app, from Unity build through to App Store Connect and Play Console release.
- Built custom internal Slack-to-VM AI agent workflows to automate stakeholder requests, code modifications, and testing pipelines.

Oct 2024 – Oct 2025

**Fabra** — Software Engineer, Sydney

- Built features across Fabra's core SaaS product in React and TypeScript.
- Took over a departing senior engineer's workstream and shipped share-design permissions and email validation end to end, front and back.
- Wrote internal tooling that cut manual steps out of the team's day-to-day backend operations.

Feb – Oct 2024

**Bardon Design** — Junior Software Engineer, Bangkok

- Worked with the Louis Vuitton team on a gamified e-commerce experience built to lift engagement through interactive shopping.
- Built an interactive AR furniture selection app.

Summers, 2021 – 2023

**Software internships** — Auckland

- Built an AR Rube Goldberg machine level for GeoArGames, and a 2D arcade game embedded in the Yuri&Niels site.

TOOLKIT

LANGUAGES

TypeScript / JavaScript  
Python  
C#  
C++

AI & WORKFLOWS

AI Agents / LLMs  
Slack API & Webhooks  
VM Automation  
Automated Testing  
Prompt Engineering

WEB & INTEGRATIONS

React  
Stripe  
REST APIs  
ESP Integrations  
Netlify / Vercel

SHIP & OPERATE

GitHub Actions  
Git Version Control  
App Store Connect  
Google Play Console  
DigitalOcean Spaces

EDUCATION

Bachelor of Design Innovation — Victoria University, 2021–2023

OUTSIDE WORK

Robotics · 3D printing · Snowboarding · Waterpolo · Toastmasters

REFERENCES

Available on request — previous engineering leads at Fabra and Bardon Design are happy to speak.